



Cognitronics

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Executive Summary

Bringing Cognex and TrustVault to the Humanoid

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What We Are Building

Humanoid robots are proliferating. The hardware is maturing fast — Clone, 1X, Figure, Tesla Optimus, NEURA. What none of them have solved is the layer above the body: a cognitive system that learns from semantically rich, embodiment-specific data, compiles that learning into deployable policies, and gives the owner sovereign, encrypted control over the robot's experiential memory.

Cognitronics builds that layer.

The humanoid body is built by manufacturers. We do not build the hardware. We bring **Cognex** and **TrustVault** to it.

Cognex is our cognitive and data layer: a live semantic supervision system that captures human task demonstrations, annotates them in real time with intent, affordances, spatial relationships, and risk flags, and turns them into structured training data at a quality no existing pipeline produces. Cognex starts as a dataset and API product. It ends as a fully compiled VLM — perception, reasoning, and action policies embedded together, ready to deploy on any eligible humanoid.

TrustVault is the sovereign memory anchor embedded within Cognex: an owner-keyed, on-device encrypted layer that stores the robot's learned experience of its environment, its owner's routines, and its accumulated contextual knowledge. The manufacturer cannot access it. The cloud does not hold a copy. The memory belongs to the owner — mathematically, by design.

Together, Cognex and TrustVault give the humanoid something no manufacturer ships by default: a cognitive layer that learns, remembers, and belongs to its owner.

The Problem

Every major humanoid robotics lab faces the same structural bottleneck: their training pipelines are **linear by design**.

Physical Intelligence, DeepMind, and 1X all rely on in-situ teleoperation — an engineer stands next to a robot, instructs it to perform a task, and the raw sensor log becomes training data. One session. One environment. One dataset increment. There is no mechanism to multiply coverage without multiplying human operator time. This does not scale.

The second problem is depth. Raw capture shows *what* happened, not *why*. It encodes motion without intent, objects without affordances, paths without risk. A robot trained on this data can imitate — it cannot reason.

The third problem is ownership. Every humanoid on the market today runs its reasoning engine in the manufacturer's cloud. The robot learns in your home. That knowledge lives in their infrastructure. The owner holds nothing.

Cognitronics solves all three — at source, in a single pipeline.

Competitive Positioning

Cognitronics does not compete with manufacturer "Minds." It supplies the **training infrastructure** (Cognex) and **owner-sovereign memory** (TrustVault) those minds lack.

Player	Training data	Semantic capture at source	Simulation scale-out	Owner-sovereign memory
Physical Intelligence / DeepMind / 1X	Linear teleop — scales with operator hours	❌ Raw motion, no intent/affordance/risk	⚠️ Internal, undisclosed	❌ Manufacturer cloud
Clone Robotics	Telekinesis (raw sessions)	❌ No structured pipeline — Cognex wedge	❌ Not public	❌ No TrustVault layer
Figure / Tesla Optimus	Proprietary VLA (Helix / FSD-derived)	⚠️ Perception-heavy, shallow embodied semantics	⚠️ Internal sim stacks	❌ Closed, OEM-controlled
Cognitronics	1 demo × domain rand → N variants	✅ Intent, affordance, risk at capture	✅ MuJoCo scale-out (funded)	✅ TrustVault — owner-keyed, on-device

Wedge → platform: We enter through Clone Alpha — 164 DoF soft actuation, 279 pre-orders, the only body generating human-like contact physics at training fidelity. Founder-level interest is established; formal partnership follows proof. The same Cognex + TrustVault stack is architected to deploy on **Figure, Optimus, Apptronik, NEURA**, and simulation-first labs — the bet is category infrastructure, not a single-OEM plugin.

Regulatory pull (TrustVault): GDPR, EU AI Act, and EU Data Act move data sovereignty from feature to compliance requirement — especially for EU OEMs and any manufacturer shipping into European households.

The Pipeline

Step 1 — Cognex Capture: Inverting the Data Bottleneck

The human operator performs tasks using Clone Telekinesis while the Cognex semantic supervisor annotates the stream in real time: object affordances, spatial relationships, intent labels, contact states, and risk flags (fragility, spill probability, electronic proximity). The output is a richly labelled HDF5 dataset — embodiment-specific to Clone Alpha's 164 DoF, 70 IMUs, and 320 pressure sensors.

That seed then feeds a **MuJoCo domain randomisation pipeline**, generating hundreds to thousands of policy rollout variants from a single session:

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1 human demonstration × domain randomisation = N simulation variants
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Physical Intelligence's cost curve is linear. Ours is not. The asymmetry compounds over time.

Step 2 — Train the Big Brain

The Cognex dataset trains a **tritych-aware foundation model** — not three separate systems bolted together, but a unified architecture with three parallel cognitive layers baked in during training:

- **Mem1 — Live layer (VLA, ~20 Hz):** Real-time multimodal-to-motor inference
- **Mem2 — Context layer (VLM, ~1–5 Hz):** Persistent working memory — self-model, spatial position, mission state, current subtask

- **Mem3 — Reasoning layer (LLM, on-demand):** Deliberative chain-of-thought — path selection, grasp strategy, risk anticipation

This is the same layered cognitive architecture that DeepMind, Physical Intelligence ($\pi 0$), NVIDIA GR00T, and 1X are independently converging toward. Cognitronics arrives at the same conclusion — with a structurally better data source feeding it.

Step 3 — Simulate, Distil, Deploy

Compiled policies are stress-tested across the full domain randomisation space in MuJoCo. Through repeated training, Mem3's explicit deliberation compresses into Mem1/Mem2 as **implicit policy priors** — the robot stops re-deriving "spill risk on path 2" from scratch; that judgment is already embedded in the action distribution. Experience becomes instinct.

Validated policies are then deployed on Clone Alpha for real-world testing. Failures loop back to Cognex for new capture sessions, continuously refining the corpus.

Proof & Traction

Signal	Status
Cognex capture PoC	Live — semantic supervision pipeline demonstrated on teleoperation stream (intent, affordances, risk flags, structured export)
Demo assets	PoC video + structured HDF5 export schema validated on Clone Alpha embodiment specs (164 DoF, 70 IMUs, 320 pressure sensors)
Simulation path	MuJoCo domain-randomisation architecture scoped; scale-out benchmark is a funded deliverable (Tranche 2)
Clone Robotics (primary wedge)	Active founder-level dialogue — cofounder contacted, mutual interest confirmed; no signed agreement yet (appropriate at pre-seed). Target: data-sharing / co-development terms + preferential hardware access as pipeline results land (M9–M10)
Platform ambition	Clone Alpha is the first deploy target — anatomically unique, highest-fidelity capture surface. Architecture is designed for downstream deployment on additional OEM stacks (Figure, Tesla Optimus, Appteronik, NEURA, and simulation-first labs) via ROS2 integration and hardware-agnostic TrustVault licensing
TrustVault	Architecture defined; Jetson enclave prototype scoped for M15–M18 (Tranche 3)

What this raise proves: structured Cognex capture produces a **measured learning delta** on the first wedge platform, with an integration path that generalises — not a one-robot custom build.

PoC demo available in dataroom.

The Products :

Product 1 — Scene-Graph & Dataset API (*immediate revenue*)

The Cognex structured dataset — scene graphs, affordance labels, intent annotations, risk flags — is available via API to any humanoid manufacturer, simulation lab, or embodied AI company that needs high-quality, semantically rich training data without the

infrastructure to generate it. This is the first commercial output of the pipeline, with near-zero marginal cost per additional consumer once the dataset exists.

Product 2 — TrustVault Sovereign Memory Layer (*SaaS / licensing*)

TrustVault is hardware-agnostic and licensable to any humanoid manufacturer as an owner-facing feature. It gives end users cryptographic ownership of their robot's experiential memory — GDPR-native by architecture, not by policy. Manufacturers integrate TrustVault as the user trust and data sovereignty layer. Owners pay for the key that is mathematically theirs. This is a recurring SaaS line with strong regulatory tailwind from GDPR, the EU AI Act, and the EU Data Act.

Product 3 — S4T0R1 as a Full Humanoid OS (*the end product*)

The natural upsell and long-term product: Cognex with compiled ACT policies, the full triptych cognitive stack, and TrustVault embedded — packaged as a deployable cognitive OS for any eligible humanoid. Not a replacement for the manufacturer's reasoning engine. A sovereign layer that sits above it: perception, memory, and action guidance that belongs to the owner. This is the humanoid OS. The API is the on-ramp to it.

The Clone Robotics Partnership

Primary wedge — not exclusive forever.

Clone Alpha is our first capture and deploy surface: 164 DoF, soft actuation, the most anatomically accurate training hardware available. We have opened a founder-level dialogue with Clone Robotics; interest is mutual, though no agreement is signed yet — logical at this stage. The near-term goal is a data-sharing / co-development arrangement: Cognitronics generates structured embodied datasets and policy benchmarks; Clone accelerates GTM on a body that uniquely benefits from them.

This is not a permanent cage. Cognex capture is embodiment-specific; TrustVault and the Scene-Graph API are not. Once the pipeline is proven on Clone, the same integration path targets **Figure**, **Tesla Optimus**, **Appteronik**, **NEURA**, and simulation-first labs — any OEM building humanoids for homes and services that needs better training data and a sovereignty layer they cannot build without surrendering their data advantage.

Each platform accelerates the other. Clone gives us fidelity; the category gives us scale.

The Moat

The dataset is not portable. A policy trained on 164 DoF soft-actuated motion cannot be naively ported to a 42 DoF rigid motor system. The action space, proprioceptive signals, and contact physics are categorically different. The data was generated through a specific body. It encodes that body's way of being in space.

No other lab captures at this embodiment fidelity. No other pipeline produces semantic annotations at source. No other product gives the owner cryptographic sovereignty over the robot's learned memory. Each is a moat. Together, they are a stack.

Business Model

Tier	Product	Model	Timing
1	Cognex — Scene-Graph & Dataset API	API licensing, per-call or subscription	Immediate — dataset exists at PoC
2	TrustVault — Sovereign Memory Layer	SaaS per robot / OEM licensing	Mid-term — pushed to manufacturers
3	S4TOR1 — Full Humanoid OS (Cognex + ACT/policies + TrustVault)	Platform licensing / humanoid OS	Long-term — natural upsell from API

One pipeline. Three revenue lines. Marginal cost decreases as the dataset and policy corpus grow.

Execution Plan — 18 Months

Raise: €750K pre-seed SAFE (rolling close) · **Target Seed:** €20–50M contingent on proof milestones

Phase	Window	Deliverable	Proof output
1 — Capture	M0–M3	Live semantic annotation prototype; teleoperation → structured dataset pipeline	Annotated task corpus
2 — Structure	M3–M6	Scene-graph + affordance generation; information retrieval over embodied sessions	Structured dataset pipeline
3 — Simulate	M6–M9	MuJoCo domain randomisation; controlled task-variation testing	Simulation variant set
4 — Benchmark	M9–M12	Learning delta vs. raw-teleop baseline (error rate, sample efficiency, robustness)	Measured learning delta
5 — Deploy	M12–M15	ROS2 integration layer; runtime schemas; policy deployment hooks on humanoid hardware	Deployable integration path
6 — Expand	M15–M18	TrustVault context runtime; owner-controlled memory; Jetson encrypted enclave demo	Seed-ready proof package

Capital → evidence mapping

Tranche	Amount	Unlocks
1	€200K	Working Cognex capture prototype
2	+€250K (€450K)	Simulated policy scaling benchmark
3	+€300K (€750K)	ROS2 path + TrustVault prototype + OEM validation pipeline

Each tranche converts capital into auditable proof — not headcount for its own sake.

The Market

Every humanoid manufacturer building for households, healthcare, and services needs structured embodied training data and a sovereignty layer — and none can generate it on hardware that isn't anatomically accurate enough to make it meaningful. The regulatory environment is actively creating demand for TrustVault. Full market analysis available in dataroom.

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